Failure mode measured on the VCS	VCS measured evidence proven in this benchmark	Neural-network analogue documented, not proven here	Status here → there
1 **Gradient vanishes on discrete / index outputs**	**0.235** gradient bucket F (position) single record The NAIVE position gradient is provably zero at a strobe-timed argmax (§3.2); the whole gradient bucket stays low, 0.235, on the position regime.	Shattered / saturated gradients; non-differentiable argmax & index readouts.	proven here analogue there
2 **Saliency is model-invariant (sanity-check fails)**	**0.000** sanity-check margin single record Program-randomization: saliency cosine = 1.000 while 113/128 RAM bytes changed -> sanity check FAILS.	Saliency unchanged under model- & label-randomization; behaves like an edge detector.	proven here analogue there
3 **Correlation ≠ causation (Granger & tuning trap)**	**0.136** Granger F 95% CI [0.09, 0.18] Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.213.	Spurious Granger edges; 'present ≠ used' tuning & attention read as explanation.	proven here analogue there
4 **Single-unit ablation has an interaction blind-spot**	**0.207** connectome F 95% CI [0.13, 0.29] Single-unit lesion is faithful (0.965) yet misses the wiring: connectome recovered at only 0.207.	Single-unit ablation misleading; computation is distributed across many units.	proven here analogue there
5 **SAE/probe: present-not-used**	**0.240** selectivity F 95% CI [0.15, 0.34] SAEs match a variable, yet score 0.153 on causal use; probes decode at 0.953 acc but 81 cells are not causal.	Control-task selectivity gap; superposition / polysemantic neurons.	proven here analogue there
6 **IG/EG baseline sensitivity**	**0.175** EG F (all-regime) 95% CI [0.12, 0.24] IG magnitude is baseline-sensitive (up to 1.000 across games); a content-byte baseline kills IG; EG lands at 0.175.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input.	proven here analogue there